

Cycling to School: Increasing Secondary School Enrollment for **Girls** in India

- Karthik Muralidharan¹, Nishith Prakash² [Article Link](#)

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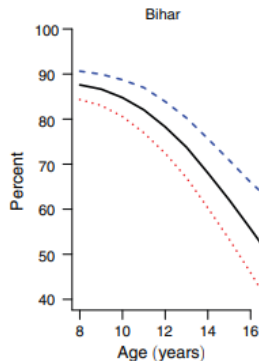
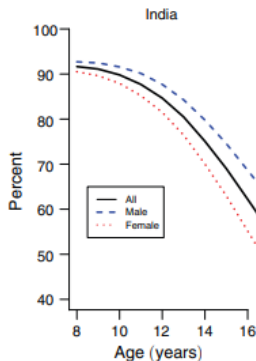
Introduction

- **Bihar** → the 3rd most populous state of India,
 - faced significant economic & educational challenges
- It had among **the lowest mean levels of education**, particularly **for girls**
- ① **Dropout Rates:** Girls' enrollment ↓ significantly ~ the ages of 14 → 15, which overlaps with the age of puberty & the transition to secondary schooling .



Introduction

Panel A. Enrollment in school by age and gender

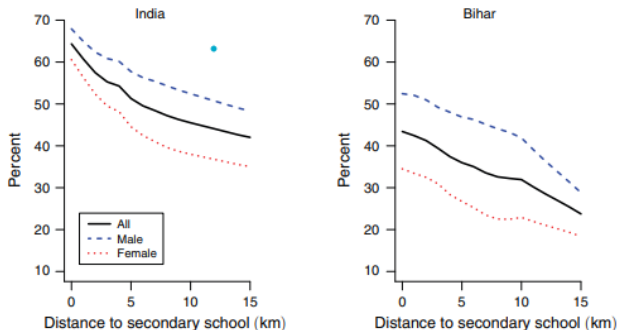


- Gender Gap in Edu.:** gender gap (Bihar) $\geq$ gender gap (national average).



Introduction

Panel B: 16- and 17-year-olds enrolled in or completed grade 9 by distance and gender



- **Distance Barrier:** ~ 90% of villages in Bihar had a primary school³, But $\leq 12\%$ of them had a secondary school.

³ 2007 DLHS

CM Bicycle Program

- The Govt. of Bihar launched the **CM's Bicycle Program in 2006**, (from here on **the Cycle Programme or CP**)
 - **AIM:** improve access to secondary education by providing **free bicycles** to **girls enrolling in grade 9**.
- ① **Conditional Kind Transfer(CKT) & Conditional Cash Transfer (CCT):**
 - CP was designed as CKT & serves as both a **demand-side & supply-side** intervention.
 - The enrollment conditionality is analogous to **demand-side CCT**, while also addressing **supply-side** barriers by providing bicycles to reduce the time, distance, and safety costs associated with attending school.
- ② **Funding:** The govt. **disbursed Rs. 2,000 per girl student** to purchase bicycles.



Implementation & Challenges

- ⁴ The CP was **well-implemented**; only a mere 3% of eligible HH not receiving funds or not purchasing bicycles.
- Factors Contributing to Success:
 - 1 **Universal eligibility:** every girl in 9th grade was entitled to the bicycle grant, which removed officials' discretion in determining beneficiaries.
 - 2 **one-time transfer:** made it easier to monitor than smaller regular transfers.
 - 3 **public ceremonies:** Funds distributed at public school ceremonies created social pressure on parents to use the cash for bicycles, with principals collecting receipts for accountability.
 - 4 **political commitment:** contributed to low leakage & successful implementation.

⁴Ghatak et al. (2016)



Implementation and Challenges

- **Causal Impact** : did CP directly caused more girls to enroll in secondary school ?
- **Economic Growth & Education Spending** :
The large post-2006 government spending on education increases in girls' secondary school enrollment may have occurred irrespective of the CP !



Objective

① Impact On Enrollment

- Changes in **NER** Of Girl Student in Bihar for grade 9 ?

② Heterogeneity Of Enrollment Impact

- Effect of **Distance** on NER & Deciphering the **dominant mechanism**

③ Impact On Learning Outcomes

- Impact on Grade 10 exam **Appearance & Passing**



Type of education	Class	Age (in yrs)
Secondary	IX	14-15
	X	15-16
Senior Secondary	XI	16-17
	XII	17-18

Table: Education vs Age



1 Indian District Level Health Survey (DLHS-3) [2007 - 2008] :

- **Sample Size:** ~ 720,000 households across 601 districts in India.
- **Content:**
 - 1 HH socioeconomic characteristics
 - 2 roster of HH members with their education attainment
 - 3 current schooling status
- **Village-level Questionnaire:** Contains data on
 - available educational facilities in the village
 - **distances** to the nearest primary, middle, & secondary schools.
 - timing makes it Ideal for our analysis

Cohort	Age (in yrs)
Treated	14-15
Control	16-17

Table: Treatment & Control Cohort



- Dataset: Households with **at least one member aged 14-17** in **Bihar & Jharkhand**

② School Level Secondary School Enrollment Data [2002 - 06] :

- **Purpose:** Test for `parallel trends` in the growth rate of enrollment of boys & girls in Bihar & Jharkhand.
- **Limitation:** Not used for estimating `treatment effects`

③ Official Data on Secondary School Certificate (SSC) Examination:

- **Content:** Provides the # of students who `appeared in` & `passed the` SSC exam.
- **Purpose:** Used to study the impact on learning outcomes.
- **Reliability:** Far more than the school-based enrollment records !



Methodology I

① 1st Difference:

- Diff (2nd ary enrollment of **Girls aged** $\underbrace{14-15 \text{ yrs}}_{\text{Treatment Cohort}}$, $\underbrace{16-17 \text{ yrs}}_{\text{Control Cohort}}$) in Bihar
- **Concern:** Possible **confounders** changing within Bihar during the program rollout.
- we control some of the confounders we introduce:

② 2nd Difference:

- Control Group for boys of the same age in Bihar
- **Rationale:** Boys were exposed to the same Non Gender Specific changes as girls but except for the Programme !
- Implement DD
- Check for Parallel Trends for 4 yrs from AY: 2002-2003 to 2005-2006
 - H_0 : There is Parallel Trends



TABLE 1—TESTING THE PARALLEL TRENDS ASSUMPTION

Dependent variable: log (9th grade enrollment by school, gender, and year)	
<i>Panel A. Testing parallel trends for the difference-in-differences (DD)</i>	
Female × year	0.0518 (0.00)
Female	−0.870 (0.06)
Year (coded as 1 to 4)	0.0852 (0.01)
Constant	4.235 (0.05)
Observations	20,266
R^2	0.167

Figure: Parallel Trends for DD

- **Reject H_0**
- Why ?? Plausible Reasons ??



- Any Solution ??

③ 3rd Difference :

- Compare the DD (Bihar, Jharkhand) $\Rightarrow$ **Triple Diff Method**
- Failed to reject H_0



Methodology IV

Panel B. Testing parallel trends for the triple differences (DDD)

Female $\times$ year $\times$ Bihar	-0.0100 (0.01)
Female $\times$ year	0.0618 (0.01)
Female $\times$ Bihar	0.175 (0.11)
Bihar $\times$ year	0.0290 (0.01)
Female	-1.045 (0.09)
Year (coded as 1 to 4)	0.0562 (0.01)
Bihar	-0.123 (0.12)
Constant	4.358 (0.11)
Observations	22,279
R^2	0.171

Figure: Parallel Trends for DDD



1. Triple Diff Estimate

Triple-Difference Estimate

$$y_{ihv} = \beta_0 + \boxed{\beta_1} F_{ihv} T_{ihv} (BH)_{ihv} + \beta_2 F_{ihv} (BH)_{ihv} + \beta_3 T_{ihv} (BH)_{ihv} + \beta_4 F_{ihv} T_{ihv} + \beta_5 F_{ihv} + \beta_6 T_{ihv} + \beta_7 (BH)_{ihv} + \epsilon_{ihv} \quad (1)$$

① $y_{ihv} \rightarrow$ Outcome Var. of interest; child i , household h & village v

② $F_{ihv} = \begin{cases} 1 & \text{if female} \\ 0 & \text{otherwise (ie male)} \end{cases}$

③ $T_{ihv} = \begin{cases} 1 & \text{"Treated" Cohort (aged 14 or 15)} \\ 0 & \text{otherwise (ie "Control" Cohort)} \end{cases}$

④ $(BH)_{ihv} = \begin{cases} 1 & \text{from Bihar} \\ 0 & \text{otherwise (ie Jharkhand)} \end{cases}$

2. Quadruple Diff Estimate

- Why ?? : Unique State Level dynamics
- Demand Side vs Supply Side effect
- Impact of Distance

Quadruple-Difference Estimate

$$y_{ihv} = \beta_0 + \boxed{\beta_1} F_{ihv} T_{ihv} (BH)_{ihv} (LD)_v + \sum_{i=2}^{i=5} \text{(4 Triple Interactions)} + \sum_{i=6}^{i=11} \text{(6 Double Interactions)} + \sum_{i=12}^{i=15} \text{(4 Linear Terms)} + \epsilon_{ihv} \quad (2)$$

① y_{ihv}

② F_{ihv}

③ T_{ihv}

④ $(BH)_{ihv}$

⑤ $(LD)_v = \begin{cases} 1 & \text{if the nearest SS} \geq 3\text{km from } v \\ 0 & \text{otherwise} \end{cases}$

3. A Regression Similar to (1)

Estimating impact on Learning Outcomes

- Use SSC data for Bihar & Jharkhand
- Consider $\log(\text{abs. \# of Students who attempted \& Passed the } 10^{\text{th}})$
- Applying a Similar DDD regression as (1)



1. Enrollment Impact

TABLE 2—TRIPLE DIFFERENCE (DDD) ESTIMATE OF THE IMPACT OF BEING EXPOSED TO THE CYCLE PROGRAM ON GIRL'S SECONDARY SCHOOL ENROLLMENT

Dependent variable: Enrolled in or completed grade 9				
Treatment group = age 14 and 15				
Control group = age 16 and 17	(1)	(2)	(3)	(4)
Treat × female × Bihar	0.103 (0.030)	0.091 (0.029)	0.052 (0.025)	0.052 (0.025)
Treat × female	0.020 (0.026)	0.024 (0.026)	0.038 (0.021)	0.039 (0.021)
Treat × Bihar	-0.044 (0.018)	-0.042 (0.018)	-0.029 (0.016)	-0.028 (0.016)
Female × Bihar	-0.094 (0.023)	-0.091 (0.023)	-0.067 (0.020)	-0.066 (0.020)
Treat	-0.148 (0.014)	-0.143 (0.014)	-0.138 (0.013)	-0.138 (0.013)
Female	-0.092 (0.020)	-0.088 (0.020)	-0.100 (0.017)	-0.101 (0.017)
Bihar	0.011 (0.016)	-0.044 (0.016)	-0.032 (0.015)	-0.044 (0.015)
Constant	0.464 (0.013)	0.771 (0.024)	0.593 (0.027)	0.562 (0.040)
Demographic controls	No	Yes	Yes	Yes
HH socioeconomic controls	No	No	Yes	Yes
Village level controls	No	No	No	Yes
Observations	30,295	30,295	30,147	30,112
R ²	0.035	0.088	0.207	0.208

Figure: DDD Results



Enrollment Impact

- Secondary School **NER** of **girls in Bihar** ↑ by 5.2% pts.
 - It represents a 32% ↑ in real terms. [16.3% → 48.3%]
- Bridged the **gender gap** in **NER** by 13% pts
 - a 40% ↑ in real terms



2. Heterogeneity of Enrollment Impact

TABLE 3—QUADRUPLE DIFFERENCE (DDDD) ESTIMATE OF THE IMPACT OF BEING EXPOSED TO THE CYCLE PROGRAM ON GIRL'S SECONDARY SCHOOL ENROLLMENT BY DISTANCE TO SECONDARY SCHOOL

Dependent variable: Enrolled in or completed grade 9				
Treatment group = age 14 and 15				
Control group = age 16 and 17	(1)	(2)	(3)	(4)
Treat × female × Bihar × long distance indicator	0.094 (0.058)	0.088 (0.056)	0.088 (0.050)	0.087 (0.050)
Treat × female × long distance indicator	−0.079 (0.050)	−0.080 (0.048)	−0.074 (0.043)	−0.073 (0.043)
Treat × female × Bihar	0.043 (0.041)	0.034 (0.039)	−0.005 (0.038)	−0.004 (0.038)
Female × Bihar × long distance indicator	−0.083 (0.045)	−0.075 (0.043)	−0.069 (0.039)	−0.070 (0.039)
Treat × Bihar × long distance indicator	−0.029 (0.036)	−0.025 (0.036)	−0.009 (0.033)	−0.009 (0.033)
Treat × female	0.072 (0.035)	0.077 (0.033)	0.088 (0.032)	0.087 (0.032)
Treat × long distance indicator	0.037 (0.029)	0.039 (0.029)	0.032 (0.026)	0.031 (0.026)
Treat × Bihar	−0.023 (0.027)	−0.022 (0.027)	−0.018 (0.025)	−0.018 (0.025)
Female × long distance indicator	0.065 (0.038)	0.063 (0.037)	0.058 (0.033)	0.057 (0.033)
Female × Bihar	−0.042 (0.032)	−0.043 (0.031)	−0.023 (0.029)	−0.022 (0.029)
Bihar × long distance indicator	0.014 (0.034)	0.022 (0.032)	0.007 (0.028)	0.008 (0.028)
Treat	−0.172 (0.023)	−0.168 (0.022)	−0.159 (0.021)	−0.159 (0.021)
Female	−0.135 (0.028)	−0.130 (0.026)	−0.138 (0.025)	−0.139 (0.025)
Bihar	−0.009 (0.026)	−0.066 (0.024)	−0.043 (0.021)	−0.054 (0.021)
Long distance indicator	−0.075 (0.028)	−0.073 (0.026)	−0.044 (0.023)	−0.040 (0.023)
Constant	0.513 (0.023)	0.816 (0.028)	0.622 (0.030)	0.587 (0.042)
Demographic controls	No	Yes	Yes	Yes
HH socioeconomic controls	No	No	Yes	Yes
Village level controls	No	No	No	Yes
Observations	30,295	30,295	30,147	30,112
R ²	0.039	0.091	0.208	0.209

Figure: DDDD Results



Heterogeneity of Enrollment Impact

- for $(LD)_v = 1$; the cycle programme $\uparrow$ the **NER of girls in Bihar** by 8.75% pts.
 - 87% $\uparrow$ in real terms. [10.1% $\rightarrow$ 18.8%]
- for $(LD)_v = 1$ **NER gender Gap ratio** $\downarrow$ by 16.3% pts.
 - 54% $\downarrow$ in real terms
- for $(LD)_v = 0$ the impact is negligible



• **Dominant Mechanism:** Reduction in the “distance cost” of attending school (and the resulting improvement in school access)



Heterogeneity of Enrollment Impact

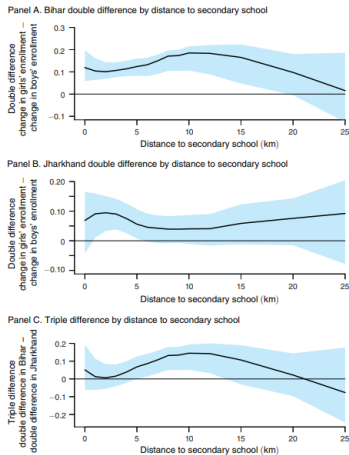


FIGURE 2. NON-PARAMETRIC DOUBLE AND TRIPLE DIFFERENCE ESTIMATES OF IMPACT OF THE CYCLE PROGRAM
(by distance to nearest secondary school)

Figure: Non Parametric DD & DDD Estimates

Heterogeneity of Enrollment Impact

- **Panel B:** Jharkhand the Catch up in NER has been consistently +ve
- **Panel A:** In Bihar too the catch up has been significant
- **Panel C:** The Bootstrapped 95% CI is +ve in **dist** $\in (5km \rightarrow 13km)$
 - in Intermediate range the effect of the Intervention is Considerable
 - **Inverted U hypothesis**



3. Impact on Learning Outcomes

TABLE 4—IMPACT OF EXPOSURE TO THE CYCLE PROGRAM ON GIRLS' APPEARANCE IN AND PERFORMANCE ON GRADE 10 BOARD EXAMS

Triple difference (DDD) estimate of impact of exposure to cycle program		
Dependent variable	log (number of candidates who appeared for the 10th grade exam) (1)	log (number of candidates who passed the 10th grade exam) (2)
Bihar × female × post	0.184 (0.065)	0.122 (0.068)
Female × Bihar	−0.266 (0.046)	−0.224 (0.047)
Bihar × post	0.083 (0.045)	0.021 (0.047)
Female × post	0.138 (0.056)	0.117 (0.058)
Female	−0.628 (0.039)	−0.697 (0.040)
Bihar	0.239 (0.032)	0.213 (0.032)
Post	0.247 (0.040)	0.171 (0.041)
Constant	4.510 (0.028)	4.280 (0.028)
Observations	15,694	15,630
R ²	0.175	0.156

Figure: Impact on Grade 10 Exam (Appearance & Passing)



Impact on Learning Outcomes

Batch	Years
"Post" Programme	2009 - 10
"Pre" Programme	2004 - 07

Table: Treatment & Control Cohort

- 18.6% pts ↑ in **Appearance**
 - In real terms 56% of the 32% increase in female secondary NER in Bihar
- 12.2% pts ↑ in **Passing**
 - In real terms 37% of the 32%
- Gains in **Learning** have been **Modest**
- ↑ in **Human Capital**



Robustness

- **Defn. of Robustness:** The capacity of research findings to maintain consistency & validity across varying conditions.
- **Significance:** Essential for establishing the reliability & credibility of research outcomes.
- **Overview:** Exploration of methodologies to address limitations & ensure robustness in our analysis.



Data Limitations & Strategies for Ensuring Robustness

- **Data Limitation:** The limitation posed by single cross-sectional data & the lack of panel data confines the analysis solely to age-appropriate girls. Issues arise regarding the under-representation of older children in households due to migration.
- **Age Distribution Analysis:** Model age of being enrolled in grade 9 is 14-15 (treated cohort), but considerable enrollment grade from age 13-16 and even 17 in Bihar lead us to underestimate the overall program impact.
- **Sample Size Examination:** The author further investigated the age distribution of respondents in the sample, noting a significant ↓ in the sample size for the 17-year-old cohort. ⇒ individuals in this cohort may have departed the household either for marriage or employment, leading to an understatement of the gender gap in this cohort.



Alternative Cohort Definitions & Results

- **Exploring various cohort definitions to validate & reinforce analysis outcomes.**
 - Author included 13 year old in treated cohort & compare cohort aged 13-15 (treated cohort) to 16-17 (control cohort).
 - Dropped 17 year old cohort from estimation & compare cohort aged 14-15 (treated cohort) to 16 (control cohort).
 - Finally, we compare cohort aged 13-15 (treatment) to one aged 16 (control cohort).
- **Results**
 - Magnitude of impact is larger when the 17-year-old group is excluded.
 - Gender gap in secondary school enrollment in Bihar is significantly higher than that of Jharkhand.
 - Significant reduction in gender gap in younger cohort who were exposed to cycle program.



Evaluation of Influencing Factors & Addressing Concerns: Placebo Test

- **Assessing the Impact of Infrastructure Improvements**

- 1 **Infrastructure Enhancements:** Investigating the role of improved roads and law enforcement in enhancing school participation.
- 2 **Cost Reduction Effects:** Exploring whether infrastructure improvements reduce barriers to school attendance, akin to bicycle programs.
- 3 **Influence on Analysis:** Assessing the potential impact of infrastructure improvements on the reliability of research estimates.

- **Conducting Placebo Tests to Validate Results**

- **Placebo Test Overview:** Utilizing triple difference specification to assess the impact on 8th-grade girls.
- **Rationale:** Selecting 8th-grade girls as a comparison group due to exposure to infrastructure improvements without program eligibility.
- **Findings:** Analysis outcomes indicating the absence of impact on 8th-grade enrollment, suggesting complementary effects with infrastructure improvements.



Border District and Clustering & Spillover Effects

- **Border District Analysis:** The point estimate in DDD is unchanged in case we restrict our sample to only border district in bihar & jharkhand rather than complete sample in way showing how robustness of result.
- **Spillover Concerns:** Potential indirect effects within households resulting from the launch of the program.
- **-ve Spillover Hypothesis:** If girls attend school & are therefore unable to perform household chores, boys might need to take on more household chores, potentially impacting their own schooling, which is quite unexpected in the context of the patriarchal setting of Bihar.
- **+ve Spillover Hypothesis:** The author suggests $\uparrow$ boys' enrollment due to the influence of their sisters' schooling.



Cost-Effectiveness & Discussion

- Comparing the CP with traditional CCT programmes:
 - 1 **Pakistani CCT** program (costs \$3/month) ↑ female enrollment in grades 6–8 by a mere 9%.⁵
 - 2 **Bangladeshi CCT** program (paid a stipend of \$0.64–\$1.50/month) had no impact on enrollment.⁶
- **The CP** cost \$1/month per recipient, & being exposed to it led to a 32% ↑ in female secondary school enrollment.

⁵ Chaudhury & Parajuli (2010)

⁶ Heath & Mobarak (2015)



Conclusion I

① Impact of the cycle program

- 32% ↑ in age-appropriate secondary school participation for exposed girls.
- 40% ↓ in gender gap in enrollment.

② Mechanism of Impact

- ↓ in "distance cost" of attending school driven by bicycles.
- Enrollment impacts concentrated for girls living $\geq 3\text{km}$ from school.
- ↑ in girls appearing for SSC exam $\implies$ genuine ↑ in participation.

③ Cost-Effectiveness

- **CP** is more **cost-effective** than equiv. cash transfer in boosting girls' enrollment.
- Suggests potential modifications to traditional conditional transfer programs for improved effectiveness.



4 Program Implementation

- Effective implementation despite challenges of leakage and corruption.
- Potential as a policy option for ↑ female education in other developing countries.

5 Long-Term Effects

- Beyond ↑ enrollment, it may contribute to delaying marriage & childbirth.
- $\implies$ demographic transition towards lower fertility rates & ↑ human capital investment in children.
- The CP **empowers women** .



References



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End

THANKS !

